Sampling, DT signals, system properties

Answers and worked solutions · Day 02 · Sun 25 Oct · pack D1

5. Practice

From your course files

Sample Test 1 Q1 (sample test, 5 marks). Sampling.

(a) $T_s = 1/F_s = 0.01$ s. Substitute $t = nT_s$: $x(n) = 10e^{-0.0001n}\cos(4\pi n) = 10e^{-0.0001n}$, because $\cos(4\pi n) = 1$ for every integer n (the 200 Hz tone aliases to DC, since $F_s/2 = 50$ Hz). So $x(0) = 10$, $x(1) = 9.9990$, $x(2) = 9.9980$, $x(3) = 9.9970$.

(b) The window $u(n+1) - u(n-2)$ is 1 for $-1 \le n \le 1$, where 2^{-n} gives $2, 1, 0.5$. There is a spike of height 1 at $n = 3$. So $x(n) = \{2, \underset{\uparrow}{1}, 0.5, 0, 1\}$ for $n = -1, \ldots, 3$: stems of height 2, 1, 0.5 at $n = -1, 0, 1$, nothing at $n = 2$, and 1 at $n = 3$. Note $x(2) = 0$.

Sample Test 1 Q2 (sample test, 4 marks). DT Signal Operations.

The underlined 1 is $x(0)$, so the samples sit at $n = -2, \ldots, 3$.

- a. $x(n-3)$ occupies $n = 1..6$ and $x(n+1)$ occupies $n = -3..2$. They overlap only at $n = 1, 2$: at $n = 1$, $x(-2) + x(2) = 8 + 6 = 14$; at $n = 2$, $x(-1) + x(3) = 9 + 1 = 10$. So $x(n-3) + x(n+1) = \{8, 9, 1, \underset{\uparrow}{9}, 14, 10, 1, 9, 6, 1\}$ for $n = -3, \ldots, 6$.
- b. $x(n/2) = \{8, 0, 9, 0, \underset{\uparrow}{1}, 0, 9, 0, 6, 0, 1\}$ for $n = -4, \ldots, 6$.
- c. $x(n)(-1)^n = \{8, -9, \underset{\uparrow}{1}, -9, 6, -1\}$.
- d. $E = 64 + 81 + 1 + 81 + 36 + 1 = 264$. This is an energy signal.

Sample Test 1 Q3 (sample test, 4 marks). DT System Properties.

- a. Linear and causal. The n coefficient makes it time-varying, and it is dynamic.
- b. Linear, time-invariant and causal. $\cos(0.2\pi)$ is a constant.
- c. Time-invariant only. The -1 makes it nonlinear, and $x[n+1]$ makes it non-causal.
- d. Linear and time-invariant. $x[n+1]$ makes it non-causal.
- None of the four is static: each has $y[n-k]$ feedback, so the output has memory.

T1 Q1 (tutorial).

Hint: look for a leftover π. A DT sinusoid is periodic only if $f_0 = \omega_0/2\pi$ is rational.

- a. A CT sinusoid is always periodic: $T_0 = 2\pi/5$ s.
- b. $f_0 = 5/(2\pi)$ is not rational, so not periodic.
- c. $f_0 = 1/(12\pi)$ is not rational, so not periodic.
- d. $f_1 = 1/(16\pi)$ (irrational) and $f_2 = 1/16$. The first factor is not periodic, so $x(n)$ is not periodic.
- e. $f_1 = 1/4$, $f_2 = 1/16$, $f_3 = 1/8$, all rational, so $N_1 = 4$, $N_2 = 16$, $N_3 = 8$ and $N = \mathrm{LCM}(4, 16, 8) = 16$.

T1 Q3 (tutorial).

$x_{\max} = 6.35$ and $x_{\min} = -6.35$, so the range is 12.7.

- a. $L = \dfrac{x_{\max} - x_{\min}}{\Delta} + 1 = \dfrac{12.7}{0.1} + 1 = 128$, and $b \geq \log_2 128 = 7$, so **7 bits**.
- b. $L = 12.7/0.02 + 1 = 636$, and $b \geq \log_2 636 = 9.313$, so **10 bits**.

T1 Q4 (tutorial).

$L = 2^b = 2^8 = 256$ levels over $x_{\max} - x_{\min} = 1$ V, so $\Delta = \dfrac{1}{L-1} = \dfrac{1}{255} \approx$ **0.0039 V**.
Bit rate $= bF_s = 8 \times 20 =$ **160 bits/s**. $F_{\max} = F_s/2 =$ **10 Hz**.

T2 Q1 (tutorial).

Key: S/D = static/dynamic, L/NL = linear/nonlinear, TI/TV = time-invariant/time-varying, C/NC = causal/non-causal, St/Unst = stable/unstable.

- a. **S, NL, TI, C, St.** $x(n) = 0$ gives $y(n) = 1 \neq 0$, so nonlinear. $|y| \leq 1$ always.
- b. **D, L, TI, NC, Unst.** $x(n-d) \mapsto \sum_{k=-\infty}^{n+1} x(k-d) = \sum_{m=-\infty}^{n-d+1} x(m) = y(n-d)$. It uses the future input $x(n+1)$. With $x(n) = u(n)$ (bounded by 1), $y(n) \to \infty$.
- c. **S, L, TV, C, St.** $x(n-d) \mapsto x(n-d)\cos(\omega_0 n) \neq y(n-d) = x(n-d)\cos(\omega_0(n-d))$.
- d. **D, L, TV, NC, St.** $x(n-d) \mapsto x(-n-d+2) \neq y(n-d) = x(-n+d+2)$. It needs future inputs for $n \leq 0$ (e.g. $y(0) = x(2)$).
- e. **S, NL, TI, C, St.** $|a_1 x_1 + a_2 x_2| \neq a_1|x_1| + a_2|x_2|$.
- f. **D, L, TV, NC, St.** $x(n-d) \mapsto x(2n-d) \neq y(n-d) = x(2(n-d))$. It needs future inputs for $n > 0$ (e.g. $y(1) = x(2)$).

T2 Q2 (tutorial).

Since T is time-invariant, delaying pair 2 by one sample gives $x_2(n-1) = \{\underset{\uparrow}{0}, 0, 0, 3\} \xrightarrow{T} y_2(n-1) = \{\underset{\uparrow}{0}, 0, 1, 0, 2\}$. Now $x_3(n) = \frac{1}{3}x_2(n-1)$, so if T were linear the output would have to be $\frac{1}{3}y_2(n-1) = \{\underset{\uparrow}{0}, 0, \frac{1}{3}, 0, \frac{2}{3}\}$. But the observed $y_3(n) = \{1, \underset{\uparrow}{2}, 1\}$ is different. T fails superposition, so it is **nonlinear**.

New questions

N1 (easy) [3]

a. $F_1 = 150$ Hz and $F_2 = 450$ Hz, so the Nyquist rate is **900 Hz**. Since $F_s = 500$ Hz, the 450 Hz tone aliases.

b. $\omega_1 = 2\pi(150/500) = 0.6\pi$ and $\omega_2 = 2\pi(450/500) = 1.8\pi \to 1.8\pi - 2\pi = -0.2\pi$. So

$$x(n) = 4\cos(0.6\pi n) - 2\sin(0.2\pi n).$$

c. Substitute $n = 500t$: $y(t) = 4\cos(300\pi t) - 2\sin(100\pi t)$. The 450 Hz tone reappears at 50 Hz with its sign inverted. ($F_a = 450 - 500 = -50$ Hz.)

N2 (easy) [2]

a. The frequencies are $f_1 = 0.15 = 3/20$, $f_2 = 0.225 = 9/40$ and $f_3 = 1/5$, so $N_1 = 20$, $N_2 = 40$ and $N_3 = 5$. Then $N = \mathrm{LCM}(20, 40, 5) = \mathbf{40}$. A numerical check confirms that no smaller N works.

b. $f_0 = 0.3/(2\pi)$ is irrational, so the signal is **aperiodic**.

N3 (medium) [3]

a. $L = 5/0.001 + 1 = 5001$, and $\log_2 5001 = 12.29$, so $b =$ **13 bits**.

b. $\Delta = 5/(2^{13} - 1) = 5/8191 =$ **0.610 mV**, and $|e_q| \le \Delta/2 =$ **0.305 mV**.

c. The bit rate is $13 \times 8000 =$ **104 kbit/s**. The highest recoverable frequency is $F_s/2 =$ **4 kHz**.

N4 (medium, chained) [4]

The indices are $n = -2, \ldots, 2$, with $x(0) = 3$.

a. $x(n-2)$ occupies $n = 0..4$ and $x(n+1)$ occupies $n = -3..1$. Subtract column by column:

n	-3	-2	-1	**0**	**1**	**2**	**3**	**4**
$x(n-2)$				2	-1	3	1	4
$x(n+1)$	2	-1	3	1	4			
result	-2	1	-3	1	-5	3	1	4

b. Substitute directly. The index $-2n+1$ is odd for every integer n, so only $x(\pm 1)$ and $x(3)$ can appear. $n=0$ gives $x(1)=1$; $n=1$ gives $x(-1)=-1$; $n=-1$ gives $x(3)=0$. So $x(-2n+1)=\{\underset{\uparrow}{1},-1\}$, with values only at $n=0,1$.

c. $x_e(\pm 2)=\frac{1}{2}(2+4)=3$, $x_e(\pm 1)=\frac{1}{2}(-1+1)=0$ and $x_e(0)=3$. So $x_e(n)=\{3,0,\underset{\uparrow}{3},0,3\}$. This is even-symmetric, as expected.

d. $E=4+1+9+1+16=$ **31**. It is finite and non-zero, so this is an **energy signal** with $P=0$.

N5 (medium) [3]

a. **Static, nonlinear, TI, causal, stable.** It is nonlinear because $x=0$ gives $y=1\neq 0$. It is stable because $|y|<e^{M_x}$.

b. **Dynamic, linear, TI, non-causal, stable.** Substituting $x(k-d)$ and setting $m=k-d$ gives $\sum_{m=n-d-2}^{n-d+2}x(m)=y(n-d)$, so it is TI. It is non-causal because $y(n)$ uses $x(n+1)$ and $x(n+2)$. It is stable because $|y|\leq 5M_x$.

c. **Dynamic, linear, time-varying, non-causal, stable.**

- Dynamic: $y(2)=x(4)$.
- Time-varying: delaying the input gives $x(n^2-d)$, but $y(n-d)=x((n-d)^2)$. Concrete case: with $x=\delta(n)$, $y=\delta(n)$; with $x=\delta(n-1)$, $y=\delta(n-1)+\delta(n+1)$, which is not $\delta(n-1)$.
- Non-causal: $y(2)$ needs the future input $x(4)$.
- Stable: $|y|\leq M_x$.

N6 (hard, chained) [5]

a. From Example 4, $r_{xx}(l)=\{-1,0,\underset{\uparrow}{6},0,-1\}$ for $l=-2,\ldots,2$, and zero elsewhere.

b. $y(n)=\{0.5,\ 1,\ \ -0.5\}$ at $n=3,4,5$. Shift x right by l, multiply and add:

- $l=1$: the overlap is $y(3)x(2)=0.5(-1)=-0.5$.

- $l = 3$: full overlap,
  $y(3)x(0) + y(4)x(1) + y(5)x(2) = 0.5(1) + 1(2) + (-0.5)(-1) = 3$.
- $l = 5$: $y(5)x(0) = -0.5$.
- Every other lag gives 0 (for example, at $l = 2$: $y(3)x(1) + y(4)x(2) = 1 - 1 = 0$).

So $r_{yx}(l) = -0.5,\ 0,\ 3,\ 0,\ -0.5$ for $l = 1, \ldots, 5$, and zero elsewhere.

c. The peak magnitude is at $\hat{d} = \mathbf{3} = D$. Also $0.5\, r_{xx}(l - 3)$ gives $-0.5, 0, 3, 0, -0.5$ at $l = 1..5$, which matches. (With the other order, $r_{xy}(l) = r_{yx}(-l)$ peaks at $l = -3$. This is why you must state your definition.)

d. From $d = 2F_s r/c$: $r = dc/(2F_s) = 3(3 \times 10^8)/(2 \times 10^6) = \mathbf{450\ m}$.

6. Quick recall check

Question

1. No. $f_0 = 0.7/2\pi$ is irrational.
2. At 300 Hz, because $1300 - 1000 = 300$ Hz.
3. 9 bits, because $2^8 = 256 < 300 \leq 512 = 2^9$.
4. It is not linear ($x = 0$ gives $y = 2$). It is time-invariant (there is no explicit n).
5. The signal energy E_x.